Yuanye Lin

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Research interests: galaxy formation and chemical evolution; galactic gas flows; environmental effects; Galactic dust mapping and interstellar-medium kinematics.

Education

Ph.D. in Astronomy, Shanghai Jiao Tong University 2021 – present
Advisor: Prof. Ying Zu; Zhiyuan Honors Ph.D. Program.

B.Sc. in Physics, Zhiyuan College, Shanghai Jiao Tong University 2017 – 2021

Research Experience

Visiting Graduate Student, University of Utah, USA Sep. 2025 – Sep. 2026
Hosts: Profs. Zheng Zheng and Kyle Dawson, Department of Physics and Astronomy.

Doctoral research, Shanghai Jiao Tong University. Research assistant role within the Zhiyuan Honors Ph.D. Program.

Grants and Fellowships

Principal Investigator, NSFC Young Student Basic Research Program May 2024 – Dec. 2026
Doctoral student program. Total funding: RMB 300,000.

"Reconstructing the Chemical Evolution of Galaxies across the Cosmic Web with DESI and JUST." Project registration no. 123B240.

T. D. Lee Ph.D. Fellowship, Shanghai Jiao Tong University 2023 – 2025

Publications and Manuscripts

Refereed publication

1. **Lin, Y.**, & Zu, Y. 2023, "Constraints on galactic outflows from the metallicity–stellar mass–SFR relation of EAGLE simulation and SDSS galaxies," *MNRAS*, **521**, 411–432. doi:10.1093/mnras/stad502.

Submitted; public preprint available

2. **Lin, Y.**, Zu, Y., et al., "Satellite Metallicity Enhancement I: Stellar Mass Loss, Suppressed Star Formation, and Enriched Inflow in EAGLE and DESI Cluster Galaxies." Submitted to *ApJ*; response to referee comments in preparation. arXiv:2605.08342.[1]

Unpublished manuscripts

3. **Lin, Y.**, et al., "Galactic Extinction and Reddening from DESI Luminous Red Galaxies." Under DESI collaboration review.

4. **Lin, Y.**, Zu, Y., et al., "Satellite Metallicity Enhancement II: Stellar-Mass and Halo-Mass Dependence of Satellite Metallicity Enhancement in DESI DR1." Manuscript complete; preparing for DESI collaboration review.

[1] The linked arXiv version carries an earlier title; the title listed here is that of the current manuscript.

Selected Research Contributions

NE-CEM: non-equilibrium chemical evolution of galaxies

- Formulated the model assumptions and derived the evolution equations, building on the model concept and EAGLE-to-SDSS strategy proposed by Ying Zu.
- Developed a power-law–exponential ("powexp") description of star formation histories and an empirical outflow mass-loading prescription using EAGLE; constructed a nine-parameter model to reconstruct its fundamental metallicity relation (FMR).
- Applied NE-CEM to reconstruct the SDSS FMR and used the inferred parameters to constrain the stellar mass–mass-loading relation, comparing the results with observational estimates.

Satellite metallicity enhancement (SME)

- Measured satellite metallicity enhancement using DESI data and compared it with EAGLE; constructed three simulation reference samples to disentangle physical contributions to chemical enrichment.
- Analyzed contributions from suppressed star formation, stellar mass loss, and enriched inflow within the NE-CEM framework.
- In SME II, measured the dependence on halo mass, stellar mass, and star formation rate, and interpreted these trends using the physical picture established in SME I.

Galactic foreground dust and gas with DESI luminous red galaxies

- Implemented the LRG dust-probe approach proposed by Zheng Zheng: constructed the sample, validated the method, brought photometry onto a common system, and developed the measurement pipeline.
- Used D4000 and redshift to control galaxy-population differences among sightlines and construct a Galactic reddening map.
- Made initial foreground Na I D velocity measurements across the DESI footprint. Ongoing work explores column-density constraints and spatial inference under an assumed Galactic rotation curve and kinematic model.

Collaborations and Survey Projects

- Member, Dark Energy Spectroscopic Instrument (DESI) Collaboration.
- Lead, DESI Project #419: "A Panoramic View of the Chemical Evolution History of the DESI Y1 Galaxies at $z < 0.6$."
- Contributor to the early survey design of the Jiao Tong University Spectroscopic Telescope (JUST).

Technical Skills

- **Programming and computing:** Python; HPC/cluster computing.
- **Methods:** chemical evolution modeling; spectral stacking; emission-line metallicity measurements; analysis of cosmological hydrodynamical simulations; photometric homogenization; Galactic reddening mapping; interstellar absorption-line analysis.
- **Data and simulations:** DESI, SDSS, EAGLE.
- **Languages:** Chinese (native), English (fluent).

References

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